

LAB1: CHE655

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Introduction

The code tries to verify the fundamental law that during matrix multiplication with a vector, the inner and outer products are always consistent.

Methodology

1. The results are tested for the equation $Ax = b$, where A is a matrix and x is a vector.
2. The inputs for x are taken first. It is assumed to be a row vector.

```
xlen = input(print('\n'+ "Enter the number of elements for vector x:"))
print("\n")
xlen = int(xlen)
x = np.zeros(xlen)

for i in range(xlen):
    usrinp = input(print("Enter the number at position " + str(i)+":"))
    x[i] = float(usrinp);
print(x)
```

3. While taking inputs of A the dimensionality is checked, the no of columns of A should be equal to no of elements of x . Progress to point 4 requires satisfying this criterion.

```
while 1: #DIMENSIONALITY CHECK
    Arow = input(print('\n'+ "Enter the number of rows for matrix A:"))
    Arow = int(Arow)
    Acol = input(print('\n'+ "Enter the number of columns for matrix A:"))
    Acol = int(Acol)
    if Acol == xlen:
        print('\n'+ "The dimensions of matrix A are consistent with the dimension of vector b.
You can proceed.")
        break
    else:
        print("The number of columns of A are not same as the number of rows of b. Matrix
multiplication NOT POSSIBLE. Kindly re-enter your inputs")
        print("\n")
```

4. The row vectors are printed. The inner product is calculated.

```
#INNER PRODUCT

print('\n'+ "Printing the row vectors of A: -")

for i in range(Arow):
    print('Row No '+ str(i))
```

```

print(A[i])
print("\n")

I = np.zeros(Arow)
print("\n" + 'Inner Product')

for i in range(Arow):
    for j in range(Acol):
        I[i]+=A[i][j]*x[j]

print (I)

```

5. This is followed by column operations. The column vectors are printed and outer product is calculated.

```

#OUTER PRODUCT

print("\n"+"Printing the column vectors of A: -")

for j in range(Acol):
    temp = np.zeros(Arow)
    print('Column No '+ str(j))
    for i in range(Arow):
        temp[i] = A[i][j]
    print(temp)
    print("\n")

O = np.zeros(Arow)
print("\n" + 'Outer Product')

for j in range(Acol):
    for i in range(Arow):
        O[i]+=A[i][j]*x[j]

print (O)

```

6. Finally each and every element of the inner and outer products are compared. If all are found to be zero the law is satisfied else the law is violated.

```

print("\n" + "Comparing inner and outer products")
for i in range(Arow):
    print("\n" + 'At position ' + str(i) + ':')
    print("\n" + "Outer Product Element:")
    print(O[i])
    print("\n" + "Inner Product Element:")
    print(I[i])
    print("\n" + "Difference between IP and OP Element At position " + str(i) + ':')
    print(O[i]-I[i])
    if O[i] == I[i]:
        print("\n" + "Elements at position "+ str(i) + ' are same')
        if i==Arow-1:
            print ("\n" + "\n" + "Inner and Outer Products have yielded the same results." + '\n')
            print(O)

```

```

else:
    print('\n' + "Fundammental law violated. The code needs correction")
    break

```

Results

Input:-

x = [1. 2.]

A = [[1. 2.]

[3. 4.]

[5. 6.]]

Analytical Output = [5 11 17]

Code Output:

Printing the row vectors of A: - Row No 0 [1. 2.] Row No 1 [3. 4.] Row No 2 [5. 6.] Inner Product [5. 11. 17.] Printing the column vectors of A: - Column No 0 [1. 3. 5.] Column No 1 [2. 4. 6.] Outer Product [5. 11. 17.] Comparing inner and outer products At position 0: Outer Product Element: 5.0 Inner Product Element: 5.0	Elements at position 0 are same At position 1: Outer Product Element: 11.0 Inner Product Element: 11.0 Difference between IP and OP Element At position 1: 0.0 Elements at position 1 are same At position 2: Outer Product Element: 17.0 Inner Product Element: 17.0 Difference between IP and OP Element At position 2: 0.0 Elements at position 2 are same
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Difference between IP and OP Element At position 0: 0.0	Inner and Outer Products have yielded the same results. [5. 11. 17.]
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